

REPORT OF DATA ANALYTICS ESSENTIALS

Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

SAKSHI PRIYA

Roll Number: 2410031125

Batch: 2024-28

Section: 3CSE7

Course Provider: CISCO NETWORKING ACADEMY

Certificate Issued: 15 June 2026

CANDIDATE’S DECLARATION

I, SAKSHI PRIYA, hereby declare that the report titled “DATA ANALYTICS ESSENTIALS” has been prepared by me to document the learning and practical competencies completed through the Cisco Networking Academy course. The report is based on the course completion credential and the learning areas stated in the certificate. All references used in preparing this report have been acknowledged. I affirm that the work presented here has been prepared for academic documentation and that I am responsible for the originality of the report.

Signature: _____

Student Name: SAKSHI PRIYA

Roll No.: 2410031125

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Cisco Networking Academy for providing the Data Analytics Essentials course and the opportunity to develop foundational knowledge of data analytics. The course helped me understand how data can be transformed into useful information through a structured analytics process, basic statistical techniques, data preparation, and commonly used analytics tools.

I am also thankful to the School of Computer Science and Engineering, IILM University, Greater Noida, for providing academic support and encouraging participation in skill-development activities. I express my gratitude to my teachers, family, and everyone who supported me during the completion of this learning activity.

Signature: _____

Student Name: SAKSHI PRIYA

Roll No.: 2410031125

INTERNSHIP / COURSE COMPLETION CERTIFICATE



Certificate of Course Completion

Sakshi Priya

has successfully achieved student level credential for completing the Data Analytics Essentials course.

The student was able to proficiently:

- *Explain how the data analytics process creates value from data.*
- *Explain the characteristics of data, including formats, availability and methods to acquire.*
- *Transform data using analytics tools.*
- *Analyze data using basic statistical and data preparation techniques.*
- *Complete hands-on lab using Excel, SQL, Tableau and other tools.*
- *Evaluate and share project portfolio.*



Scan to Verify

Lynn Bloomer
Lynn Bloomer
Director, Cisco Networking Academy

Issued on: Jun 15, 2026

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1. PROJECT DESCRIPTION

The DATA ANALYTICS ESSENTIALS course is a foundational learning program completed through CISCO NETWORKING ACADEMY. The course focuses on understanding the data analytics process and the ways in which raw data can be prepared, analyzed, interpreted, and communicated. The completion certificate states that the learner was able to explain how the data analytics process creates value from data; explain data characteristics such as formats, availability, and acquisition methods; transform data using analytics tools; analyze data using basic statistical and data-preparation techniques; complete hands-on lab activities using Excel, SQL, Tableau, and other tools; and evaluate and share a project portfolio.

1.1 Introduction

Data analytics is the process of working with data to identify useful information, patterns, and insights that can support better decisions. A typical analytics workflow begins with understanding the available data, followed by acquiring, preparing, transforming, analyzing, and communicating the results. This course provided a structured introduction to these activities and helped develop an understanding of the role of data quality and appropriate analytics tools in producing meaningful results.

The learning activity is relevant to a B.Tech CSE student because data analytics concepts are used in many software and artificial-intelligence applications. Understanding spreadsheets, SQL-based data handling, basic statistics, visualization, and the overall analytics process provides a useful foundation for further study in data science, machine learning, and data engineering.

1.2 Organization / Course Provider Profile

Cisco Networking Academy is the course provider identified on the completion certificate. The certificate shows that the student achieved a student-level credential for completing the Data Analytics Essentials course. The learning content represented in this report is therefore limited to the competencies explicitly indicated on the certificate rather than claiming employment or an industry job role.

1.3 Problem Statement

Organizations generate data in many formats and from different sources, but raw data is not automatically useful. A major challenge is to understand the characteristics of data, obtain it appropriately, prepare it for analysis, apply suitable analytical techniques, and communicate the resulting information clearly. Without a structured process, data may contain inconsistencies or may be difficult to interpret. The learning problem addressed by this course is to build a foundational

understanding of how data can be converted into meaningful information using standard analytics practices and tools.

1.4 Project Objectives

- Understand how the data analytics process can create value from data.
- Understand data characteristics, including formats, availability, and methods of data acquisition.
- Learn the purpose of data transformation and preparation before analysis.
- Apply basic statistical and data-preparation concepts to analyze data.
- Gain hands-on exposure to analytics tools including Excel, SQL, and Tableau.
- Understand how analytical results can be evaluated and communicated through a project portfolio.

1.5 Scope of the Project

The scope of this report covers the foundational data analytics competencies documented in the course certificate. It includes understanding the analytics lifecycle, data formats and availability, data acquisition, data transformation, basic statistical analysis, data preparation, visualization-oriented work, and the use of Excel, SQL, Tableau, and other analytics tools. The report does not claim advanced machine-learning development, production deployment, or professional employment experience because these areas are not stated on the supplied certificate.

1.6 Technologies and Tools Used

Tool / Technique	Purpose
Excel	Used as a spreadsheet-based analytics tool for working with and analyzing data.
SQL	Used for structured data handling and querying concepts.
Tableau	Used for data analytics and communicating analytical information through visual outputs.
Basic Statistical Techniques	Used for understanding and analyzing data at an introductory level.
Data Preparation Techniques	Used to organize and transform data before analysis.

1.7 System Architecture

A conceptual data analytics workflow can be represented as a sequence of stages: Data Sources → Data Acquisition → Data Preparation and Transformation → Analysis → Visualization / Communication → Evaluation and Sharing. This architecture is a conceptual representation of the analytics process described by the course competencies; it is not a claim that a particular production system was implemented.

1. Data Sources
2. Data Acquisition
3. Data Preparation and Transformation
4. Analysis
5. Visualization / Communication
6. Evaluation and Portfolio Sharing

1.8 Methodology

1. Understand the objective of the analytics task and identify the type of information required.
2. Examine the available data, including its format, availability, and possible acquisition method.
3. Prepare and transform the data so that it is suitable for analysis.
4. Apply basic statistical and analytical techniques to identify useful information.
5. Use suitable analytics tools such as Excel, SQL, and Tableau for hands-on analysis activities.
6. Evaluate the analytical output and communicate or share the results in an understandable form.

1.9 Expected Outcomes

After completing the learning activity, the expected outcome is a clear foundational understanding of the data analytics process and the ability to explain how raw data can be transformed into useful information. The learner should also be able to recognize common data characteristics, understand basic data preparation and statistical analysis, and identify appropriate tools for different analytics tasks. Exposure to Excel, SQL, and Tableau provides a practical foundation for future academic projects involving data analysis and visualization.

1.10 Certificate and Supporting Proof

The supplied certificate confirms that SAKSHI PRIYA successfully achieved a student-level credential for completing the DATA ANALYTICS ESSENTIALS course through CISCO NETWORKING ACADEMY. The certificate was issued on 15 June 2026. A copy of the certificate is attached in this report as supporting proof of completion.

2. BIBLIOGRAPHY / REFERENCES

Cisco Networking Academy. Data Analytics Essentials — Course Completion Certificate issued to Sakshi Priya, 15 June 2026.

Cisco Networking Academy. Data Analytics Essentials course learning materials and hands-on lab activities, as completed by the student.

Microsoft Excel — spreadsheet-based data analysis tool referenced in the course competency list.

SQL — structured query language referenced in the course competency list.

Tableau — analytics and visualization tool referenced in the course competency list.